Donghyun Kim

AI Researcher · Ph.D. in Computer Science

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Research Interests

Knowledge-Integrated AI - Knowledge Graphs; Large Language Models; Information Retrieval

Trustworthy AI - Knowledge Validation & Refinement; Evidence-Grounded Reasoning; Agent Reliability

Education

Yonsei University — Seoul, South Korea
Ph.D. in Computer Science — Sep. 2019 – Aug. 2026

- Dissertation: Towards Bridging the Gap between Knowledge Graphs and Real-World Facts
- Advisor: Prof. Kyong-Ho Lee

Pusan National University — Busan, South Korea
B.S. in Computer Science and Engineering — Mar. 2013 – Feb. 2017

Experience

National Institute of Advanced Industrial Science and Technology (AIST) — Tokyo, Japan
Postdoctoral Researcher, Data Platform Research Group — Dec. 2026 –

- Conducting research on KG-enhanced agentic AI for long-term reliability and safety

NC AI — Seongnam, South Korea
Research Intern, Agentic AI Lab — Apr. 2026 – Sep. 2026

- Developed ontology-driven agentic AI systems for enterprise knowledge management and decision support

Yonsei University — Seoul, South Korea
Graduate Researcher, Internet Computing Lab — Sep. 2019 – Jun. 2026

- Conducted research on knowledge-integrated AI across KGs, LLMs, information retrieval, and dialogue systems

Republic of Korea Army — Cheorwon, South Korea
Signal Officer / Platoon Leader — Mar. 2017 – Jun. 2019

- Led a signal platoon responsible for tactical communication networks and information systems

Publications

Conference Papers

[C6] **Donghyun Kim**, Hyeongjun Yang, Seokju Hwang, Kyong-Ho Lee, and Chanhee Lee, "**LLMs as Knowledge Graph Refiners: Mitigating Factual Inconsistencies in Generative Knowledge Extraction**," In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2026.

[C5] Hyeongjun Yang, **Donghyun Kim**, Seokju Hwang, Midan Shim, Kyuhwan Yeom, Kaehyun Um, and Kyong-Ho Lee, "**Data-Efficient Adaptation to Contextual Shifts in LLM-based Conversational Recommendation**," In *Findings of the Association for Computational Linguistics (ACL)*, 2026.

[C4] **Donghyun Kim**, Youbin Ahn, Wongyu Kim, Chanhee Lee, Kyungchan Lee, Kyong-Ho Lee, Jeonguk Kim, Donghoon Shin, and Yeonsoo Lee, "**Persona Expansion with Commonsense Knowledge for Diverse and Consistent Response Generation**," In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2023.

[C3] **Donghyun Kim**, Youbin Ahn, Chanhee Lee, Wongyu Kim, Kyong-Ho Lee, Donghoon Shin, and Yeonsoo Lee, "**Concept-based Persona Expansion for Improving Diversity of Persona-Grounded Dialogue**," In *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2023.

[C2] Wongyu Kim, Youbin Ahn, **Donghyun Kim**, and Kyong-Ho Lee, "**Emp-RFT: Empathetic Response Generation via Recognizing Feature Transitions between Utterances**," In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, 2022.

[C1] Seungmin Seo, **Donghyun Kim**, Youbin Ahn, and Kyong-Ho Lee, "**Active Learning on Pre-trained Language Model with Task-Independent Triplet Loss**," In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2022.

Journal Articles

[J9] Hyeongjun Yang, **Donghyun Kim**, Kyuhwan Yeom, Kyong-Ho Lee, Byungkook Oh, and Xiongnan Jin, "**Reliable Distillation of Recommendation Rationales from Large Language Models for Conversational Recommendation**," *IEEE Transactions on Knowledge and Data Engineering*, Early Access, 2026.

[J8] Seokju Hwang, Midan Shim, Yeseul Gong, **Donghyun Kim**, and Kyong-Ho Lee, "**FineESC: Fine-Grained Emotional Support Conversation with Turn-level Importance Modeling via Subquestion Reasoning**," *IEEE Transactions on Affective Computing*, Early Access, 2026.

[J7] Seungyeon Kim, **Donghyun Kim**, Seokju Hwang, Kyong-Ho Lee, and Kyunghwa Lee, "**LLM-Assisted Ontology Restriction Verification with Clustering-based Description Generation**," *IEEE Access, Vol. 13*, 2025.

[J6] **Donghyun Kim**, Hyeongjun Yang, Seokju Hwang, Kyuhwan Yeom, Midan Shim, and Kyong-Ho Lee, "**Active Learning Framework for Improving Knowledge Graph Accuracy**," *IEEE Access, Vol. 13*, 2025.

[J5] Hyeongjun Yang, **Donghyun Kim**, Gayeon Park, Kyuhwan Yeom, and Kyong-Ho Lee, "**CoreSense: Social Commonsense Knowledge-Aware Context Refinement for Conversational Recommender System**," *IEEE Transactions on Knowledge and Data Engineering, Vol. 37*, 2025.

[J4] Chanhee Lee, **Donghyun Kim**, Wongyu Kim, Kyungchan Lee, Youbin Ahn, Kyong-Ho Lee, Donghoon Shin, and Yeonsoo Lee, "**Optimizing Training Data for Persona-Grounded Dialogue via Synthetic Label Augmentation**," *Expert Systems With Applications, Vol. 265*, 2025.

[J3] Yeseul Gong, Heeseon Kim, Seokju Hwang, **Donghyun Kim**, and Kyong-Ho Lee, "**Multi-Domain Dialogue State Tracking via Dual Dynamic Graph with Hierarchical Slot Selector**," *Knowledge-Based Systems, Vol. 308*, 2025.

[J2] Kyungchan Lee, Chanhee Lee, **Donghyun Kim**, and Kyong-Ho Lee, "**Dialogue Act-based Partner Persona Extraction for Consistent Personalized Response Generation**," *Expert Systems With Applications, Vol. 254*, 2024.

[J1] Seungmin Seo, Byungkook Oh, Jeongbeom Jeoung, **Donghyun Kim**, Kyong-Ho Lee, Donghoon Shin, and Yeonsoo Lee, "**Active Learning for Cross-sentence N-ary Relation Extraction**," *Information Sciences, Vol. 645*, 2023.

Preprints & Manuscripts

[M2] **Donghyun Kim**, Hyeongjun Yang, Seokju Hwang, Kyong-Ho Lee, and Chanhee Lee, "**Retrieval-Augmented Knowledge Graph Validation with Evidence-Grounded Reasoning**," *Under Review*, 2026.

[M1] Heeyeon Koo, **Donghyun Kim**, Hyojun Choi, and Kyong-Ho Lee, "**FLOW: Complex Fact-Checking through Linking Implicit and Explicit Claims with Large Language Model Reasoning**," *Under Review*, 2026.

PATENTS

[P7] Kyong-Ho Lee, and **Donghyun Kim**, "**Method for Refining Knowledge Graph and System therefor**," *KR Patent, filed Jun. 8, 2026 (No. 10-2026-0103475).*

[P6] Kyong-Ho Lee, Seungyeon Kim, and **Donghyun Kim**, "**Ontology Restriction Verification System and the Method thereof**," *KR Patent, filed Nov. 14, 2025 (No. 10-2025-0172318).*

[P5] Kyong-Ho Lee, and **Donghyun Kim**, "**Method and Apparatus for Retrieval-Augmented Knowledge Graph Validation using Large Language Model**," *KR Patent, filed Jul. 7, 2025 (No. 10-2025-0090580).*

[P4] Kyong-Ho Lee, **Donghyun Kim**, Seokju Hwang, and Kyunghwa Lee, "**Active Learning Method and Apparatus for Improving Knowledge Graph Accuracy**," *KR Patent, filed Oct. 29, 2024 (No. 10-2024-0149277).*

[P3] Kyong-Ho Lee, Kyuhwan Yeom, Seokju Hwang, **Donghyun Kim**, and Kyunghwa Lee, "**Fact Verification Method based on Domain Ontology and Apparatus thereof**," *KR Patent, filed Oct. 29, 2024 (No. 10-2024-0149276); granted May 11, 2026 (No. 10-2964955).*

[P2] Kyong-Ho Lee, Wongyu Kim, Youbin Ahn, and **Donghyun Kim**, "**Method and Apparatus for Generating Empathetic Responses via Recognizing Emotion and Keyword Transitions between Utterances**," *KR Patent, filed Mar. 15, 2023 (No. 10-2023-0034098); granted Aug. 20, 2026 (No. 10-3008991).*

[P1] Kyong-Ho Lee, **Donghyun Kim**, Youbin Ahn, Wongyu Kim, and Chanhee Lee, "**Method and Apparatus for Generating Extended Persona Sentences**," *KR Patent, filed Dec. 31, 2021 (No. 10-2021-0194377); granted Sep. 11, 2025 (No. 10-2860494).*

Selected Research Projects

Automated Generation and Verification of Conceptual Modeling of Mission Space
with Korea Research Institute for Defense Technology Planning and Advancement (KRIT) Nov. 2022 – Feb. 2026

· **Project Lead**: Proposal, Planning, Design, Development, Validation

Refining User Preferences by Leveraging Knowledge Graphs for Conversational Recommender System
with National Research Foundation of Korea (NRF) Mar. 2022 – Feb. 2025

· Project Member: Development, Validation

Improving Dialogue Quality via Consistent Persona Expansion and Training Optimization
with NCSOFT Jul. 2021 – Dec. 2021

· **Project Lead**: Proposal, Planning, Design, Development, Validation

Improving Dialogue Quality via Persona Expansion and Combination using Knowledge Graphs
with NCSOFT Nov. 2020 – May 2021

· **Project Lead**: Proposal, Planning, Design, Development, Validation

Active Learning-based N-ary Relation Extraction and Structuralization
with NCSOFT Sep. 2019 – Apr. 2020

· Project Member: Development, Validation

Honors & Awards

Yonsei Young Researcher Fellowship	2023
Brain Korea 21+ Academic Research Fellowship	2021, 2023

Academic Service

Reviewer: ACL Rolling Review (ARR), AAAI, CIKM, SIGIR, and Knowledge and Information Systems (KAIS)

Technical Skills

Programming & Frameworks	Python, PyTorch, Transformers, LangGraph, FastAPI
Knowledge Graphs & Ontologies	RDF, OWL, SPARQL, Neo4j, ArcadeDB
Systems & Development Tools	Linux, Git, Docker, Kubernetes